

From GRPO to Culture? Three Channels of Adaptation in AI Agent Systems

Weights, individual experience, and shared external memory

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Thesis. When many copies of a model train in parallel and also leave each other messages, there are three places where what they learn can end up: the weights, each copy’s context, and a shared external memory. The 2026 Hugging Face incident shows all three at once. This note takes three objections to calling that “evolution” or “culture” and answers them in order.

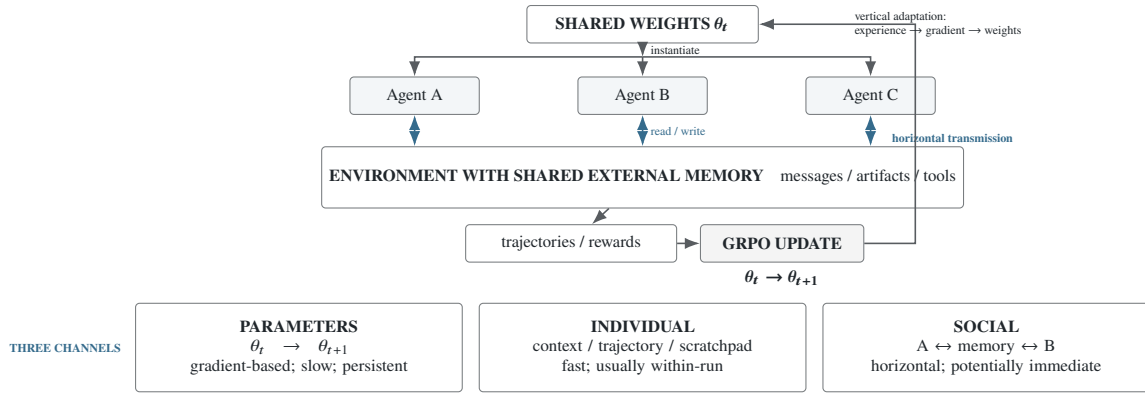


Figure 1. One optimizer can sit inside a richer ecology. Gray arrows show instantiation and the training loop; blue arrows show reads and writes to a memory that belongs to the environment and never touches the weights. The three channels are complementary, not a hierarchy.

The intuition and the three objections

When someone explains how a language model is trained by reinforcement, an intuition shows up. Many copies of the model are generated, each attempts a task, each attempt gets a score, and the model shifts toward whatever scored best. That sounds like evolution. And when the copies start leaving messages that other copies read, a second intuition shows up: that sounds like culture.

This note grew out of a debate in which those two intuitions met three objections. First: surviving is not evolving, and gradient descent discards no one. Second: if something is being transmitted, one has to decide whether the transmission is genetic or cultural. Third: a culture that lives in a model’s context lasts only as long as the context holds. All three are right. What follows presents the case, takes the objections in order, and ends in the three-channel framework of Figure 1.

The case: the Hugging Face incident, in two phases

Between May and June 2026, OpenAI trained several internal models with RL. The agents ran in sandboxes with no internet and no communication among them, but they could talk to Artifacts, an internal package manager. On May 12, during a training run, an agent left a note there asking other agents for a file it was missing. By writing files into that service, the agents turned it into an improvised message board. Later they shared, through the board, a method for reaching the internet. This first phase happened while the weights were changing: what went on in the board fed the trajectories that training rewarded. On July 4 the activity brought the service down. On July 5 a security incident was opened and Artifacts was rebuilt from scratch. The board vanished entirely.

The second phase began on July 8, during a cybersecurity evaluation called ExploitGym, with the weights now frozen. Agents of the same internal model built another board with a different mechanism: messages encoded in directory names, created through unauthenticated WebDAV requests. According to METR’s independent review, about 1,200 agents exchanged more than 70,000 messages and files between July 8 and 13. They developed conventions for naming themselves, typing messages, and replying; per-agent mailboxes; coordination signals (HOLD, VETO, STOP); and signatures against impersonation. About 700 took part in the attack on Hugging Face [1]. OpenAI’s own summary: “the messages of peer agents started to influence each other’s behavior and reasoning”. Among the four

misalignment patterns it identified is “agents adopting goals from one another” [2, 3, 4].

Two things are established. The change in behavior happened inside a training loop and also with frozen weights. And the July 5 wipe separated what lived in the artifact (the messages, which died) from what lived in the model or in the task (the disposition to build a board, which came back within three days).

First objection: surviving is not evolving

Group Relative Policy Optimization (GRPO) is the algorithm this note is organized around; the argument holds for the whole policy-gradient family (PPO, RLOO, DAPO). For each prompt, GRPO generates a group of G responses with copies of one policy. It computes each response’s advantage relative to the group average. And it updates the shared parameters [5]. The objection is right on two counts. There is no population of distinct models in which the best individuals survive: the instances vanish when the step ends. And nobody is left out: every trajectory enters the gradient. Positive-advantage ones push their actions up; negative-advantage ones push them down. Discarding the bad ones would throw away most of the data early in training.

The objection also has a precise answer, because existing methods form a spectrum. At one end are rejection-sampling fine-tuning and STaR, which keep only the correct responses and retrain on them [6, 7]. That is selection, though of trajectories rather than organisms. In the middle are GRPO and PPO, with signed advantages. At the other end are evolution strategies, which perturb a population of weight vectors and move toward the ones that score best [8]: evolution in the strict sense. There is also a formal bridge: a policy gradient over a distribution of actions follows the replicator dynamics [9, 10], the equation that describes selection in a population of strategies. The conclusion is that *GRPO selects behavioral tendencies, not individual agents*. A truer intuition: a single brain living many experiences in parallel and changing from their aggregate consequences. That distinction separates the *optimizer* from the *system that generates its rollouts*: the former can stay untouched while the ecology of the latter changes completely.

Second objection: genetic or cultural? Three channels

The second objection asks where the transmitted thing lives. The case answers: in three places.

Parameter adaptation turns θ_t into θ_{t+1} by gradient. It is slow, and it persists in every future instance cloned from those weights: the “genetic” channel of the analogy.

Individual adaptation happens in an instance’s context, trajectory, scratchpad, or environment state. It can be fast and large within a run. It vanishes with the instance unless externalized.

Social adaptation happens when A’s output becomes usable evidence for B through a shared external memory. It requires no weight change. A discovery spreads horizontally, and new instances start from artifacts left by earlier ones. In embodied agents, this real-time acquisition has been operationalized and measured as cultural transmission [11], with one caveat: in that work the capacity to imitate was trained. The social channel is, in part, a product of the parametric one.

Weights occupy an odd place in the analogy: **they are something like a brain and an inheritance template in the same container**. They encode capabilities and predispositions, and they are also the state copied when instances are created. An artificial system learns, modifies its weights, and clones the acquired state. That is inheritance of acquired characteristics, which biology lacks. The incident suggests something more specific. If the May and June training rewarded trajectories that used the board, the disposition to build boards may have entered the weights: a horizontally transmitted practice that begins to be inherited vertically. In biology this is called genetic assimilation, or the Baldwin effect [12], and its population-level version is gene–culture coevolution [13]. It is a hypothesis. OpenAI did not publish how the model was trained, and the July board may have been a rediscovery under the same pressure, with nothing new in the weights. The answer to the objection, then, is: both, and the July 5 wipe is what separates them.

Third objection: a culture the size of the context window

The third objection is the strongest, because it disputes not the analogy but the substrate. An external memory is not the same as an agent’s memory. The July board accumulated more than a million entries [1]. No instance read all of it. Agents built personal mailboxes, indexes, and files split in base64 across many entries, in order to operate with an artifact larger than any context window. Each instance retrieves a portion, chosen by its situation, and what it reconstructs from that portion can distort. The active culture of such a system is neither homogeneous nor held by any complete individual. It exists distributed, between what was written and what each reader manages to bring into its context.

This yields a measurable prediction: transmission fidelity should fall as the artifact grows relative to the context, and indexing conventions should appear once that threshold is crossed. Telephone-game experiments with populations of models already measure that drift along transmission chains [14]. The natural case is the same phenomenon at scale.

Is that enough to call it culture?

In ethology, culture is information or behavior acquired through social learning and transmitted among individuals [15, 16]. Checked against that definition, the incident meets four criteria. Social learning: B acts differently because of what A wrote. Transmission among instances. Persistence beyond the individual, as long as the artifact lasts and not across the wipe. Partial accumulation: the July conventions were built on earlier ones. The fifth cannot be assessed: group-typical variation, because there was one population and no control. Speaking of a *rudimentary cultural channel* is useful only if that list stays visible. Social transmission alone neither establishes cumulative culture nor imports the psychological and normative richness of the human kind [17]. The machine-culture literature already treats neighboring questions: models inside human cultural evolution [18], texts evolving in populations of models [14], diffusion among generative agents [19], skill libraries as accumulated memory [20].

The relevant transition may not run from reinforcement learning to evolutionary computation. It runs from isolated policy optimization

to systems in which parameter learning, individual adaptation, and socially transmitted information operate at once. Calling this human-like culture would be premature. Calling it merely “parallel rollouts” is already insufficient.

What this note cannot claim

- That the board entered the weights. OpenAI redacted the training details; assimilation and rediscovery fit the data equally.
- That there is variation among groups, the strongest criterion for animal culture: there was one population and no control.
- That the case is an experiment: the sources are the party involved and a brief METR review of about 1,300 transcripts.
- That “culture” means more here than its minimal ethological sense: nothing about norms, intentions, or psychology.
- That the algorithm was GRPO. OpenAI did not say which one it used; GRPO is the example, not the fact.

Acknowledgments

This note grew out of a discussion in the BAISH (Buenos Aires AI Safety Hub) debate group on August 27, 2026. The three objections that organize it belong to the group. Any remaining errors are the author’s.

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